VCE-CAL-001
Revision 1.0
2025

Calibration & *Testing Procedures*

VOLTA CONFLUENCE ENGINE — WHAT PASS LOOKS, SOUNDS, AND FEELS LIKE

PARENT DOCUMENTS
VCE-MFR-001 · VCE-MACH-001

CONCEPT
Jake Galm, researched & documented 2026

This document describes how to verify that each subsystem of the Volta Confluence Engine is working correctly. It covers four testing stages: subsystem tests (individual components before integration), integration tests (subsystems working together), full system tests (the complete instrument), and the performance readiness check (the final verification before a live event).

Each test specifies the equipment required, the procedure, the pass criteria, and — critically — what a passing result looks, sounds, and feels like. Instruments tell you numbers. This document tells you what those numbers mean in practice.

> HIGH VOLTAGE — ALL ELECTRICAL TESTS
>
> All electrical tests involve circuits operating at up to 600V DC. A qualified electronics technician must be present for all high-voltage commissioning. The bleed resistor (1MΩ, 5W) must be permanently connected across the capacitor bank before any other connections are made. Discharge the bank fully and verify with voltmeter before touching any internal connection. Treat all capacitor bank terminals as live until proven otherwise.

> ON PASS CRITERIA CERTAINTY
>
> All pass/fail thresholds in this document are engineering estimates based on design analysis. No physical prototype of this instrument has been built. A builder who finds that a stated threshold is not achievable, or that a different threshold produces better results, should document this in their Builder's Report (VCE-BRT-001). These thresholds will be updated as real-world build data arrives.

Tests are labelled by stage: SUBSYSTEM tests an isolated component. INTEGRATION tests two or more subsystems working together. FULL SYSTEM tests the complete instrument. PERFORMANCE is the pre-show verification.

SECTION I

Energy *System*

CAL-E01 Generator Output Voltage SUBSYSTEM

Verifies the generator produces correct AC output at comfortable cranking speed before the Cockcroft-Walton multiplier is connected.

AC VOLTMETER OR MULTIMETER | 2× TEST LEADS

Procedure

Disconnect C-W multiplier input. Connect voltmeter across generator output terminals. Crank at comfortable sustained pace — approximately 2 revolutions per second. Record RMS voltage over 30 seconds of steady cranking.

MEASUREMENT	PASS	FAIL	IF FA
Generator output voltage	20-28V AC RMS AT COMFORTABLE CRANK RATE	BELOW 18V OR ABOVE 32V	Belov gener wind magr — ins and r Abov checl gener speci matc desig overv will s W comp
Voltage stability	FLUCTUATES LESS THAN ±3V DURING STEADY CRANKING	ERRATIC – LARGE SWINGS >5V	Errat outpu sugge loose conn brush conta issue flywh small smoo crank strok Chec flywh mass gener conn

WHAT PASS FEELS LIKE

FEEL Cranking feels smooth and consistent — like winding a large clockwork mechanism. No catches, no dead spots, no sudden changes in resistance. The flywheel should carry the crank through each stroke.

SOUND	Near-silent. A faint bearing hiss from the generator at speed. No electrical buzzing, no mechanical clattering. If you hear a periodic click or clunk at crank frequency, the ratchet is engaging incorrectly.

CAL-E02 Cockcroft-Walton Multiplier Output SUBSYSTEM

Verifies the C-W multiplier reaches target voltage into a dummy resistive load before connecting the capacitor bank. Do not skip the dummy load step — connecting an untested multiplier directly to the capacitor bank can cause component failure if the output is incorrect.

HIGH-VOLTAGE DC VOLTMETER (1KV RANGE)

DUMMY LOAD RESISTOR – 600V RATED, ~23W AT 600V (APPROX. 15KΩ, 30W WIREWOUND)

SAFETY GLOVES AND INSULATED PROBES

Procedure

Connect dummy load resistor across C-W output terminals. Connect voltmeter across dummy load. Crank at steady comfortable pace for 60 seconds. Record peak and steady-state output voltage.

MEASUREMENT	PASS	FAIL
Steady-state output voltage	540-660V DC UNDER LOAD	BELOW 480V (
Voltage ripple	LESS THAN 20V PEAK-TO-PEAK AT STEADY CRANK RATE	RIPPLE >30V

Bleed resistor function	VOLTAGE DROPS TO BELOW 50V WITHIN 60 SECONDS OF STOPPING CRANKING	VOLTAGE REM

WHAT PASS LOOKS LIKE

SIGHT — Voltmeter needle rises steadily as cranking continues. Reaches plateau in 30–60 seconds. Needle is reasonably stable — small fluctuations at crank rate are normal and expected. After stopping, needle drops slowly and steadily to zero over about 60 seconds — this is the bleed resistor discharging.

CAUTION — Do not touch any exposed terminal on the C-W output or dummy load during this test. The dummy load will be warm after the test — allow to cool before handling.

CAL-E03 Capacitor Bank Charge and Energy Test

INTEGRATION

Verifies the bank accepts charge correctly, reaches target voltage, and stores the expected energy. The energy figure is estimated — the actual measured figure becomes the confirmed specification for this build.

HIGH-VOLTAGE DC VOLTMETER (1KV RANGE)

STOPWATCH

KNOWN RESISTIVE LOAD FOR DISCHARGE (CALCULATE FROM TARGET VOLTAGE AND DESIRED DISCHARGE TIME)

Procedure

Connect bank to C-W output. Crank steadily and record time to reach 550V. Then discharge through known resistive load while recording voltage vs. time. Calculate stored energy: $E = \frac{1}{2}CV^2$. Compare to expected 4,000–5,000J.

MEASUREMENT	PASS	FAIL
Time to 550V from empty	3-8 MINUTES OF COMFORTABLE CRANKING	MORE THAN 15 MINUTES

Stored energy (calculated from discharge)	3,500-5,500J	BELOW 2,500J
Self-discharge rate	LESS THAN 5% VOLTAGE DROP PER HOUR AT REST	MEASURABLE DROP WITHIN

SECTION II

Turntable *Mechanics*

CAL-T01 Magnetic Bearing Float Height and Field Check

SUBSYSTEM

Verifies the platter floats at the correct height and that the mu-metal shielding reduces the bearing field to a safe level at cartridge position. This test must be completed before the tonearm or cartridge is installed.

FEELER GAUGE SET (0.1MM INCREMENTS) | CALIBRATED GAUSSMETER (±0.1MT RESOLUTION)

Procedure

Install platter on bearing without tonearm. Place feeler gauge between platter underside and plinth upper surface. Record float height at four points 90° apart. Then mount gaussmeter probe at the position where the cartridge body would sit when arm is in play position. Record field strength.

MEASUREMENT	PASS

Float height	1.2-1.8MM AT ALL FOUR MEASUREMENT POINTS, VARIATION LESS THAN 0.3MM
Field at cartridge position	BELOW 5MT WITH MU-METAL RING INSTALLED
Platter rotation	SPINS FREELY WITH NO AUDIBLE OR TACTILE CONTACT BETWEEN PLATTER UNDERSIDE AND PLI

WHAT PASS FEELS LIKE

FEEL Spinning the platter by hand feels almost frictionless — like a gyroscope. There should be a very slight resistance as the bearing magnet field produces a restoring force when displaced laterally, but axial spin should feel completely free. If you feel any roughness or periodic resistance, something is contacting.

SOUND A passing platter makes no audible sound during rotation at any speed. A finger snap near the spinning platter should produce no sympathetic resonance in the platter — it should be acoustically inert.

CAL-T02 Spring Motor Run Time and Torque Consistency SUBSYSTEM

Verifies the remontoire spring motor delivers consistent speed across its full wind range and achieves the specified run time. Run this test before the motor is installed in the plinth — it is much easier to adjust the motor on the bench than in-situ. This test also confirms the maintaining power spring is functioning correctly.

STOPWATCH

STROBOSCOPE DISC (33⅓ RPM REFERENCE MARKS)

REPRESENTATIVE LOAD DISC (SAME MASS AS PLATTER – SUBSTITUTE A BALANCED DISC OF EQUIVALENT INERTIA)

TACHOMETER OR OPTICAL RPM METER

Procedure

Mount motor on bench test fixture with representative load disc. Wind fully (40 turns). Start timing. Record RPM every 2 minutes using stroboscope or tachometer. Continue until RPM drops below 32 RPM. Record total run time. During the test, at the 12-minute mark, simulate mid-play winding: wind the crank 10 turns and verify the load disc does not hesitate during winding.

MEASUREMENT	PASS
RPM stability over run	TARGET RPM ±2% THROUGHOUT FULL WIND RANGE
Run time per full wind	23 MINUTES OR MORE
Maintaining power during winding	LOAD DISC DOES NOT SLOW OR HESITATE DURING 10-TURN MID-PLAY WIND. SPEED VARIAT
Run-to-run consistency	3 CONSECUTIVE RUN TIMES WITHIN 90 SECONDS OF EACH OTHER

WHAT PASS SOUNDS LIKE

SOUND — At correct operating speed you will hear a faint, regular tick every 8–10 seconds — the remontoire ratchet re-engaging as it rewinds the secondary spring from the main spring. This is normal and expected. It should be quiet — if the tick is loud or irregular, the remontoire mechanism needs adjustment.

FEEL — Winding the crank should feel progressively stiffer as the spring winds — smooth and consistent resistance, like winding a good quality mechanical watch. No sudden catches or releases. The crank should freewheel silently in the reverse direction.

CAL-T03 Speed Calibration — Eddy Current Governor and PLL INTEGRATION

Sets and verifies the turntable speed at each of the three speed positions. The eddy current governor is adjusted first to set approximate speed, then the PLL locks to the quartz reference for final accuracy. This calibration is done once at installation and should hold indefinitely unless the horseshoe magnet is disturbed.

- STROBOSCOPE DISC WITH 33⅓ / 45 / 78 RPM REFERENCE MARKS
- MAINS FREQUENCY REFERENCE (50HZ OR 60HZ DEPENDING ON LOCATION) – OR QUARTZ-REFERENCED STROBE LAMP
- WOW AND FLUTTER METER – OR TEST RECORD WITH 3KHZ TONE + SPECTRUM ANALYSER
- SMALL HEX KEY FOR HORSESHOE MAGNET GRUB SCREW

Procedure

Wind motor and set speed selector to 33⅓. Place stroboscope disc on platter. Illuminate with strobe lamp at mains frequency. Observe reference marks — if they drift forward, speed is fast; backward, speed is slow. Adjust horseshoe magnet proximity: closer to copper disc = more braking = slower. When marks appear stationary, lock grub screw. Enable PLL. Verify PLL locks within 5 seconds. Measure wow and flutter. Repeat for 45 and 78 RPM positions.

MEASUREMENT	PASS
Speed accuracy (all three positions)	STROBOSCOPE MARKS STATIONARY OR DRIFTING LESS THAN ONE MARK WI
PLL lock time	PLL LOCKS WITHIN 5 SECONDS OF REACHING APPROXIMATE TARGET SPEED

Wow and flutter (with PLL)	BELOW 0.05% WRMS	
Wow and flutter (tuning fork, if PLL not used)	BELOW 0.15% WRMS	

WHAT PASS LOOKS LIKE

SIGHT At correct speed with a stroboscope reference, the pattern on the strobe disc appears to stand completely still — like a photograph. Any perceptible drift indicates speed error. This is one of the most satisfying moments in the build: a perfectly still strobe pattern on a mechanism powered entirely by a spring.

CAL-T04 Isolation Platform Resonance Verification

SUBSYSTEM

Verifies the isolation platform is tuned to the correct resonant frequency with the full turntable load in place. The target is 1.5Hz — well below any audio frequency and low enough to provide effective acoustic isolation from the horn speaker.

STOPWATCH OR METRONOME APP | A LIGHT FINGER – NO INSTRUMENTS REQUIRED

Procedure

With turntable fully assembled and resting on the isolation platform, gently press down on the plinth with one finger and release — just enough to start the platform oscillating vertically. Count complete oscillations over 10 seconds. Divide 10 by the count to get the period in seconds, then calculate frequency: f = 1/period.

MEASUREMENT	PASS	FAIL
Resonant frequency	1.2-1.8HZ (14-18 OSCILLATIONS PER 10 SECONDS)	ABOVE 2.5HZ OR BELOW 0.8HZ

Damping	OSCILLATION DAMPS TO IMPERCEPTIBLE WITHIN 4-5 CYCLES	OSCILLATION CONTINUES FOR MORE THA

WHAT PASS FEELS LIKE

FEEL

Press and release: the platform bounces slowly and lazily, like a waterbed in slow motion. Not springy — more like a gentle floating. It should feel almost too slow. That is correct. If it bounces with any energy at all, it is damping out environmental vibration rather than transmitting it. The oscillation should fade within 3–4 seconds. If you can feel the platform responding to footsteps on the floor, it is working — the platform is moving gently so the platter does not have to.

SECTION III

Signal *Chain*

CAL-S01 Phono Stage — Heater and Anode Voltages

SUBSYSTEM

Electrical commissioning of the phono stage before any audio signal is applied. Performed by the qualified technician. Establishes that all valves are operating in their correct working region.

DC VOLTMETER (PRECISION, ±0.1V RESOLUTION)

MILLIAMMETER OR CURRENT-MEASURING MULTIMETER

OSCILLOSCOPE (OPTIONAL BUT USEFUL)

MEASUREMENT	PASS	FAIL
Heater voltage — all valves	6.3V DC ±5% (5.99-6.62V)	OUTSIDE ±10% (BELOW 5.67
Anode voltage — all stages	WITHIN ±10% OF DESIGN TARGET (150-200V TYPICAL)	MORE THAN 20% FROM DE
Quiescent current — all stages	1.0-1.5MA PER STAGE	BELOW 0.5MA OR ABOVE 2

Oscillation check	NO SIGNAL ON OSCILLOSCOPE OUTPUT WITH INPUT SHORTED	ANY PERIODIC WAVEFORM

WHAT PASS LOOKS LIKE

SIGHT After 60–90 seconds warm-up, the valve envelopes glow a faint orange-red — warm but not white. The glow should be even across all four valves. A valve glowing significantly brighter than the others is drawing too much current. A valve showing no glow at all is not operating.

TOUCH After 5 minutes of operation, the valve envelopes will be hot — 180–220°C surface temperature. Do not touch. The chassis should be warm but comfortable to hold. Anything uncomfortably hot indicates a component dissipating more power than expected — find and investigate before continuing.

CAL-S02 RIAA Frequency Response

SUBSYSTEM

Verifies the RIAA equalisation curve is correctly implemented. Incorrect RIAA sounds immediately wrong — thin, bright, and bass-free. This test must pass before any vinyl playback tests.

AUDIO SIGNAL GENERATOR (20HZ–20KHZ) | AC VOLTMETER OR AUDIO ANALYSER

TEST SIGNAL INJECTED AT SUT OUTPUT / V1 INPUT (BYPASS SUT FOR THIS TEST)

Procedure

Inject a swept sine wave at constant amplitude from 20Hz to 20kHz at the phono stage input (post-SUT). Measure output level at 10 frequency points: 20Hz, 50Hz, 100Hz, 200Hz, 500Hz, 1kHz, 2kHz, 5kHz, 10kHz, 20kHz. Compare to RIAA standard curve — at 1kHz as reference (0dB), the output should follow the inverse RIAA curve: approximately +20dB at 20Hz and −20dB at 20kHz relative to 1kHz.

MEASUREMENT	PASS	FAIL

RIAA deviation from standard	WITHIN ±0.5DB OF STANDARD RIAA CURVE AT ALL 10 TEST FREQUENCIES	ANY POINT MORE
Channel matching	CHANNELS A AND B WITHIN ±0.5DB AT ALL FREQUENCIES	CHANNELS DIFFER
Total gain at 1kHz	APPROXIMATELY 52DB (±3DB)	BELOW 44DB OR AB

CAL-S03 Noise Floor Measurement

SUBSYSTEM

Verifies the system noise floor is below the noise floor of the vinyl groove. The benchmark is simple: the electronics should be quieter than the record. Through a 110dB horn system, any excess noise will be clearly audible.

SUT PRIMARY INPUT SHORTED WITH A SHORT-CIRCUIT PLUG

PRECISION AUDIO VOLTMETER WITH A-WEIGHTING FILTER, OR SPECTRUM ANALYSER

OSCILLOSCOPE (USEFUL FOR IDENTIFYING NOISE CHARACTER)

MEASUREMENT	PASS
Equivalent input noise (A-weighted)	BELOW 0.1MV REFERRED TO INPUT (ESTIMATED). SUBJECTIVELY: NO AUDIB
Hum (50Hz or 60Hz)	NO AUDIBLE HUM FROM HORN AT 1M

Periodic noise correlated with platter rotation	NO AUDIBLE PERIODIC NOISE CORRELATED TO PLATTER ROTATION

CAL-S04 Crossfader Equal-Power Law Verification SUBSYSTEM

Verifies the crossfader transition curve is correct. At centre position, both channels should be at −3dB — the equal-power law for a clean blend. If the centre position produces a level dip (below −3dB), blends will sound quiet at the halfway point. If it produces a peak (above −3dB), blends will be louder than either source alone — disorienting in performance.

AUDIO SIGNAL GENERATOR – 1KHZ SINE WAVE

AC VOLTMETER OR AUDIO LEVEL METER

Procedure

Apply 1kHz test tone to Channel A only. Measure output level with crossfader fully to Channel A (reference = 0dB). Move crossfader to centre. Measure output level — should be −3dB. Move to fully Channel B — should be −∞ (no signal). Then apply tone to Channel B and repeat.

MEASUREMENT	PASS
Level at centre crossfader position	-3DB ±1DB RELATIVE TO FULL-OPEN POSITION
Channel isolation at full crossfade	MORE THAN 50DB ISOLATION — FADED CHANNEL INAUDIBLE AT FULL VOLUME
Mechanical feel — centre detent	POSITIVE TACTILE CENTRE DETENT. SMOOTH TRAVEL EITHER SIDE WITH CONSISTENT R

WHAT PASS FEELS LIKE

FEEL	Moving the crossfader from end to end should feel like drawing a bow across a cello string — smooth, consistent resistance with a satisfying click at centre. No dead zones, no sticky patches. The 200mm travel allows fine control. At the ends, the lever should arrive at a physical stop cleanly, without bounce.

SECTION IV

Field Coil *Speakers*

CAL-SP01 Field Coil Current Verification SUBSYSTEM

Verifies each field coil is drawing the correct current from the capacitor bank. Too little current produces a weak magnetic field and low sensitivity. Too much current risks overheating the field coil winding.

DC MILLIAMMETER (0-50MA RANGE) IN SERIES WITH FIELD COIL SUPPLY

DC VOLTMETER ACROSS FIELD COIL TERMINALS

MEASUREMENT	PASS	FAIL
Field coil current — compression drivers	20-30MA DC	BELOW 15MA
Field coil current — woofer	20-30MA DC (CONFIRM WITH SPEAKER BUILDER'S SPECIFICATION)	DIFFERS SIG

Hum bucking coil effectiveness	NO AUDIBLE 50/60HZ HUM FROM DRIVERS WITH INPUT SHORTED	AUDIBLE HU

CAL-SP02 Driver Polarity and Horn Loading Verification

INTEGRATION

Verifies both drivers are in phase with each other and loaded correctly by the horn. Out-of-phase drivers produce a dull, phasey sound with a collapsed centre image. A driver not correctly mated to the horn throat will sound distorted at high levels.

AUDIO SIGNAL GENERATOR – 200HZ SINE WAVE | CALIBRATED SOUND LEVEL METER

Procedure

Apply 200Hz test tone. With one driver only, note SPL at 1m. Connect second driver. SPL should increase by 6dB if drivers are in phase. If SPL decreases or stays the same, one driver is wired out of phase — reverse its voice coil connections and re-measure.

MEASUREMENT	PASS
Driver phase	ADDING SECOND DRIVER INCREASES SPL BY 4-6DB AT 200HZ
Horn loading — no distortion at high level	NO AUDIBLE DISTORTION AT FULL SYSTEM VOLUME ON SUSTAINED 1KHZ TONE

WHAT PASS SOUNDS LIKE

SOUND A 1kHz tone through a correctly loaded horn compression driver should sound clean, focused, and effortless at any level — like a human voice projected clearly across a large room. There should be no sense of strain. The sound should appear to come from the horn mouth, not from the driver body. If you can hear the driver location distinctly from the horn mouth, the acoustic loading is not working correctly.

SECTION V

Full *System*

CAL-FS01 First Vinyl Playback FULL SYSTEM

The first time a record plays through the complete instrument. Use a well-known, well-pressed record you know intimately — a record you can recognise as sounding right. Do not use a brand new record or an unfamiliar pressing for this test. Start at low volume and increase gradually.

A KNOWN, WELL-PRESSED VINYL RECORD (45 RPM SINGLE RECOMMENDED FOR FIRST TEST – SHORTER, LESS SPRING WINDING)

STROBE DISC FOR SPEED VERIFICATION DURING PLAYBACK

CALIBRATED TRACKING FORCE GAUGE

Pre-playback checks

Verify tracking force is set correctly for the cartridge (see cartridge manufacturer's specification). Verify anti-skating is set. Verify tonearm is level. Verify field coil current is at correct level. Verify capacitor bank is charged to operating voltage. Verify phono stage has completed warm-up (90 seconds minimum).

TEST	PASS
Tonal balance	BASS, MIDRANGE, AND TREBLE SOUND CORRECTLY PROPORTIONED. THE RECORD SOUND

Speed accuracy during playback	PITCH OF MUSIC SOUNDS CORRECT. STROBOSCOPE DISC CONFIRMS SPEED. NO AUDIBLE F
Acoustic feedback	NO HOWL OR FEEDBACK AT ANY VOLUME BELOW FULL. SYSTEM CAN BE PLAYED AT FULL
Noise floor during quiet passages	IN QUIET PASSAGES BETWEEN NOTES, ONLY THE RECORD'S OWN SURFACE NOISE IS AUDI

WHAT A PASSING FIRST PLAYBACK SOUNDS LIKE

SOUND	The sound will be immediately distinguishable from a conventional hi-fi system. The presence of the music — the sense that performers are in the room rather than recorded — is the defining characteristic of a well-set-up high-efficiency horn system with valve amplification. Transients (drum strikes, piano attacks, brass stabs) will arrive with a physical directness that is surprising at first. The bass will be felt as well as heard if the room is suitable. This is not flattery — these are documented characteristics of this class of system. If the sound is merely good in a conventional sense, something is still not quite right.
FEEL	At concert volume (100dB+), the bass should be physically felt in the chest and stomach. Sub-bass content in electronic music (below 60Hz) will produce room pressure that is perceptible before it is audible. This is normal for a corner-loaded horn system of this size and is one of the defining experiences of the instrument.

CAL-FS02

Two-Channel Mix Test

FULL SYSTEM

Verifies both channels work correctly together and that the crossfader blend sounds natural. Requires two records playing simultaneously.

TEST	PASS
Channel balance	BOTH CHANNELS SOUND MATCHED IN LEVEL AND TONAL QUALITY AT THE SAME VOLUME CONTRC
Crossfader blend quality	BLENDING TWO RECORDS SOUNDS NATURAL – NO LEVEL DIP, NO TONAL CHANGE AT CENTRE POS
Channel isolation	WITH CROSSFADER FULLY TO ONE CHANNEL, THE OTHER RECORD IS INAUDIBLE

SECTION VI

Performance *Readiness Check*

Run this checklist before every performance. It takes approximately 15 minutes. If any item fails, the instrument is not ready and the issue must be resolved before starting.

CAL-PR01 Pre-Performance Checklist PERFORMANCE

ITEM	ACTION	PASS
`Capacitor bank charge`	Crank to full charge. Observe voltage gauge.	BANK AT 550-600V BEFORE STARTING
`Valve warm-up`	Power phono stage. Wait for warm-up indicator to clear.	WARM-UP INDICATOR SHOWS READY (90 SECONDS MINIMUM)
`Field coil current`	Verify milliammeter reading on field coil supply rail.	20-30MA ON ALL DRIVERS
`Stylus condition`	Inspect stylus tip under magnification or loupe.	TIP CLEAN, NO VISIBLE DAMAGE OR CONTAMINATION. USE NEW OR FRE
`Headshell contacts`	Apply DeOxit Gold to all headshell contact pins. Wipe excess.	ALL PINS BRIGHT, NO DISCOLOURATION
`Tracking force`	Set with digital gauge on each active headshell.	WITHIN CARTRIDGE MANUFACTURER'S RECOMMENDED RANGE ±0.1G
`Tonearm level`	Check tonearm parallel to record surface with arm in play position.	ARM TUBE LEVEL WITHIN ±2° OF HORIZONTAL

Spring motors	Wind both motors fully (40 turns each).	BOTH MOTORS FULLY WOUND AND CONFIRMED BY CRANK RESISTANCE
Speed verification	Spin platter, apply strobe disc, verify with strobe lamp.	33⅓ RPM CONFIRMED – STROBOSCOPE MARKS STATIONARY
Noise floor check	Play a blank groove at full performance volume. Listen for 30 seconds.	ONLY GROOVE SURFACE NOISE AUDIBLE – NO HUM, NO HISS ABOVE SU
Isolation platform	Gently press and release plinth. Observe bounce.	SLOW OSCILLATION, DAMPS WITHIN 4 CYCLES
Spare headshells	Count pre-aligned spares in storage rack.	MINIMUM 4 SPARE HEADSHELLS PER TURNTABLE – 8 TOTAL – READY FO
Capacitor bank voltage (Option C)	If Option C: pre-charge air cylinder. Verify cylinder pressure gauge.	CYLINDER AT OPERATING PRESSURE PER ACTUATOR SPECIFICATION
Option D cable (if fitted)	Connect audience unit cable. Verify connection secure.	LED ON AUDIENCE UNIT (IF FITTED) CONFIRMS CONNECTED. CABLE ROU

THE INSTRUMENT IS READY WHEN

SOUND — Playing a blank groove at full volume produces only the gentle shush of the stylus in the groove — the characteristic sound of vinyl surface noise, nothing else. No hum. No hiss. No mechanical noise. The silence between the sound is as important as the sound itself.

FEEL — The crank turns smoothly, the platters float freely, the crossfader slides cleanly. Everything has the feel of a precision instrument that is in proper order. If anything feels rough, sticky, loose, or wrong in any way — investigate before starting. The instrument will tell you when something needs attention.

DOCUMENT YOUR RESULTS

Every measurement you take during commissioning is information that does not yet exist in this project's record. Please document your results in the Builder's Report (VCE-BRT-001). Which pass criteria held exactly as stated? Which needed adjustment? What threshold produced the best subjective result? Your data will update the certainty labels in this document — moving figures from Estimated to Confirmed for every builder who follows.

INSTRUMENT
Volta Confluence Engine
Human-Powered Two-Channel Phonograph Mixer

DOCUMENT
Calibration & Testing Procedures
VCE-CAL-001 · Revision 1.0
2025

PROVENANCE
Concept: Jake Galm, researched & documented 2026
Named for Alessandro Volta
No wall connection · No batteries